

Chapter 1: Foundations of Distributed Systems

Paragraph 1. Distributed systems consist of autonomous computing entities that communicate over networks to accomplish shared goals. Fundamental challenges include synchronization, consensus algorithms, latency, and partial failures. Modern web architectures rely on replicated state machines and consensus algorithms like Raft and Paxos to ensure consistency across multiple availability zones.

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End of chapter summary and discussion points.

Chapter 2: Concurrency Models and Thread Management

Paragraph 1. Concurrency allows applications to handle multiple tasks out-of-order or in partial order without affecting the final outcome. In multi-threaded programming, mutexes, semaphores, and condition variables guard access to shared memory. Asynchronous event loops offer an alternative model where non-blocking I/O operations are coordinated on a single thread or thread pool.

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End of chapter summary and discussion points.

Chapter 3: Pipeline Architecture and Stream Processing

Paragraph 1. Data pipelines transform raw data into actionable information through a series of discrete processing stages. Each stage takes input from its predecessor and produces output for its successor. Stream processing frameworks such as Apache Kafka and Apache Flink handle unbounded streams with exactly-once semantics and watermarking mechanisms.

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Chapter 4: Microservices and Service Mesh Networks

Paragraph 1. Microservice architectures decompose monolithic applications into independently deployable units. A service mesh provides a dedicated infrastructure layer for managing service-to-service communication, offering features like load balancing, service discovery, encryption with mutual TLS, and detailed telemetry collection.

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End of chapter summary and discussion points.

Chapter 5: Relational Databases and ACID Guarantees

Paragraph 1. Relational database management systems provide atomicity, consistency, isolation, and durability. Query planners parse SQL statements into relational algebra trees and optimize execution paths using cost models and table statistics. Indexing strategies like B-trees and hash indexes dramatically reduce disk access latency.

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Chapter 6: NoSQL Data Stores and Eventual Consistency

Paragraph 1. Non-relational data stores sacrifice traditional ACID constraints to achieve horizontal scalability and high write throughput. Document stores, key-value caches, and wide-column databases model data according to access patterns. Eventual consistency guarantees that all replicas will converge given sufficient time without updates.

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End of chapter summary and discussion points.

Chapter 7: Machine Learning Inference and Accelerators

Paragraph 1. Deep learning models require specialized hardware accelerators such as GPUs and TPUs for matrix multiplications. Tensor execution runtimes optimize compute graphs through layer fusion and quantization from 32-bit floating point to 8-bit integers or 4-bit weights. Low-latency inference is critical for interactive applications.

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End of chapter summary and discussion points.

Chapter 8: Compiler Design and Intermediate Representations

Paragraph 1. Compilers translate high-level source code into efficient machine instructions. Modern optimizing compilers like LLVM use static single assignment form intermediate representations. Optimization passes include dead code elimination, loop unrolling, vectorization, and register allocation using graph coloring algorithms.

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Chapter 9: Operating System Kernels and Memory Virtualization

Paragraph 1. Operating system kernels manage hardware resources and enforce isolation between user-space processes. Virtual memory mapping relies on page tables and translation lookaside buffers. Context switching involves saving and restoring processor registers and updating memory management unit page directory pointers.

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Chapter 10: Security Protocols and Public Key Cryptography

Paragraph 1. Cryptographic protocols secure communications across untrusted networks. Transport Layer Security negotiates cipher suites and exchanges asymmetric keys using elliptic curve Diffie-Hellman algorithms. Digital certificates signed by trusted certificate authorities verify domain identity and prevent man-in-the-middle attacks.

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